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RESEARCH ARTICLE
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An LLM Framework for Regional Climate Services: Integrating Climate Knowledge and Ensemble Projections

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Key Points:

- We develop a climatology-specific large language model to support decision making for climate change adaptation
- The model holds basic knowledge about climatology and can also use local adaptation plans and future climate simulation data
- It outperforms its base model on climatology-specific benchmarks and enables quantitative planning in a case study

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Recent advances in generative AI have enabled the integration of scientific knowledge into natural language interfaces. However, existing large language models (LLMs) lack domain-specific expertise and cannot directly utilize simulation data essential for risk assessment. This study develops a climatology-specific LLM fine-tuned on climate science literature and augmented with ensemble projection data set, aiming to support local governments and small enterprises in climate adaptation planning as a preliminary support tool. The framework combines domain-specific fine-tuning, retrieval-augmented generation (RAG) for scientific documents and local guidelines, and future projection data. The developed model demonstrated highly competitive performance compared to general-purpose LLMs such as Swallow 70B on climate-specific benchmarks in both English and Japanese, particularly in the categories of "Impacts, adaptation, and vulnerability" and "Mitigation." A case study in Kumagaya City, Japan, demonstrated that the model can quantitatively update heat-mitigation measures using probabilistic projections and dynamically retrieved local constraints, generating data-informed baselines to facilitate human deliberation under multiple warming scenarios. The proposed approach bridges the gap between climate science and the early stages of adaptation planning by enabling AI-driven access to both textual and numerical climate knowledge. By lowering technical barriers to expert-level analysis, it facilitates inclusive and data-driven climate services for diverse stakeholders. This study represents the first systematic demonstration of an LLM that integrates ensemble projection data to support regionally grounded adaptation planning, contributing to the development of next-generation climate services that enhance local resilience.

Plain Language Summary Effective climate adaptation requires information that is both scientifically credible and usable by non-specialists. We develop a large language model (LLM) that holds domain knowledge in climate science and crucially can retrieve numeric values from future climate projection data. By combining access to future climate projection simulation data with text retrieval, this approach enables decision support for assessing local climate change impacts such as extreme events and for developing corresponding adaptation measures. In a proof-of-concept for Kumagaya City, it used projected temperature increases under a high-warming scenario to quantify the need for heat-mitigation measures and synthesize a feasible plan. Our model shows significant improvements over its base model and competitive performance against general-purpose LLMs on climate-focused queries. To our knowledge, this is the first demonstration of an LLM that integrates ensemble projection data to support regional climate adaptation planning.

1. Introduction

Global warming is accelerating the frequency and severity of climate-related hazards such as extreme heat, heavy rainfall, droughts, and sea-level rise. The impacts of global warming are projected to intensify further, posing serious threats to human societies and ecosystems around the world (IPCC, 2021). Addressing these challenges requires scientifically grounded assessments of future climate risks and the prompt development and implementation of appropriate adaptation strategies (IPCC, 2022a). Such measures are essential not only for reducing vulnerability to climate change but also for laying the foundation for sustainable development at both global and regional scales. Among the key stakeholders responsible for climate change adaptation, local governments play a central role in developing context-specific action plans.

However, the process of climate risk assessment and adaptation planning remains a significant challenge, particularly for actors with limited expertise or resources. Successful planning requires interdisciplinary knowledge that spans climate science, economics, industrial systems, and public policy. Many local governments,

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especially in resource-constrained contexts, face difficulties in securing the necessary financial means and human capital to perform such comprehensive analyses. As a result, there is a widening gap in adaptive capacity across different regions, which risks exacerbating existing inequalities in climate resilience (Fila et al., 2023; Fujita et al., 2023; Fünfgeld et al., 2023; Hübner & Finkbeiner, 2025). While the need for scientifically grounded climate services that are easy for non-experts to use is widely recognized, such methods have not been established, thereby limiting their usability among local governments (Lioubimtseva & da Cunha, 2023).

Recent advances in generative artificial intelligence (AI), particularly large language models (LLMs), have demonstrated the potential to facilitate access to expert knowledge through natural language interfaces; however, existing models face challenges in capturing domain-specific expertise in climatology. LLMs possess broad foundational knowledge across a wide range of disciplines and exhibit strong performance in various natural language processing tasks such as information retrieval, summarization, and reasoning (Brown et al., 2020; Chowdhery et al., 2022; Devlin et al., 2019; OpenAI, 2023). Owing to their coverage of fundamental knowledge in natural and social sciences and the ease of their use, they are increasingly regarded as a promising approach for assessing the impacts of global warming and supporting the development of countermeasures (Koldunov & Jung, 2024; Stede & Patz, 2021). Nevertheless, general-purpose LLMs such as GPT and Llama series are primarily trained on publicly available web content and therefore lack sufficient expertise in climate science (Bulian et al., 2023; Fore et al., 2024; Zhu & Tiwari, 2023). Although some studies have applied LLMs to the analysis and evaluation of climate-related texts, concerns have been raised that they may produce hallucinations with respect to scientific expertise (Chuang et al., 2025; Hsu et al., 2024). Domain-specific fine-tuning using scientific publications and technical reports, as well as the integration of external knowledge through retrieval-augmented generation (RAG; Gao et al., 2023; Lewis et al., 2020), are expected to partially mitigate these limitations (Manivannan et al., 2024; Nguyen et al., 2024). However, they cannot fully incorporate quantitative projections of future climate, which are indispensable for effective risk assessment and planning, because extracting numerical data registered in a database requires considerable technical effort. Therefore, this study develops a climate-specialized LLM capable of utilizing climate change future projection data.

The primary objective of this study is to develop an AI-based framework that enables local governments to conduct scientifically sound climate risk assessments and design adaptation strategies, particularly in environments with limited specialized expertise and resources. Specifically, we aim to construct a climatology-specific LLM that integrates domain-specific knowledge with climate projection data to support actionable decision-making. The central research question we address is: "How can AI technologies, particularly climate-specialized LLMs, be designed to facilitate robust climate risk assessment and adaptation planning for stakeholders with limited expertise and capacity?" To answer this question, we propose a comprehensive methodology that combines three key components: (a) fine-tuning the base LLM with climate science literature to enhance domain knowledge, (b) implementing RAG approaches to dynamically access external documents, including scientific reports and guidelines, and (c) integrating future projection data sets as quantitative knowledge sources within the model. This approach seeks to bridge the gap between climate science and practical decision-making through AI-enabled climate services.

2. Materials

2.1. Textual Resources

To fine-tune the base LLM, incorporate external knowledge, and evaluate the model's performance, we used a variety of text data sets. As climatology-related text data, we collected a total of 338 academic articles, primarily sourced from the Climate Change Adaptation Information Platform (A-PLAT: https://adaptation-platform.nies.go.jp/materials/ref/index.html). A-PLAT is operated by the National Institute for Environmental Studies (NIES), the central hub for climate change measures in Japan. As part of its activities, NIES researchers curate and list academic literature based on specific quality criteria; we utilized all English-language papers available in this curated list. These articles span a range of thematic fields, with the number of papers in each category as follows: "Agriculture, Forestry, Fishery" (73), "Water Environment and Resources" (15), "Natural Ecosystems" (86), "Natural Disasters and Coastal Areas" (24), "Human Health" (56), "Industrial and Economic Activities" (6), and "Life of Citizenry and Urban Life" (8). This numerical distribution is not intentional but reflects the current availability of papers in the A-PLAT database. In addition, we collected three chapters from the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report, covering Working Groups I, II, and III

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(IPCC, 2021, 2022a, 2022b). Specifically, these correspond to the physical science basis (WG I), impacts, adaptation and vulnerability (WG II), and mitigation of climate change (WG III). All of these documents are written in English and were used for fine-tuning the model to enhance its foundational knowledge in climatology and global warming and model's performance evaluation.

Furthermore, we collected climate-related articles written in Japanese and local adaptation plans published by Japanese municipalities. These documents contain detailed and practical information that can serve as valuable references when generating Japanese adaptation plans. They are utilized through RAG to support context-specific responses.

2.2. Future Projection Data

In this study, we utilized the "database for Policy Decision making for Future climate change (d4PDF)" as a data set that enables quantitative evaluation of uncertainties under future climate change. The d4PDF consists of experimental results generated using high-resolution atmospheric models with horizontal resolutions of 60 km for the global domain and 20 km for the region surrounding Japan (Ishii & Mori, 2020; Mizuta et al., 2017). Its temporal horizon encompasses a 60-year historical period (1951–2010) and two future experiments representing global mean temperature increases of approximately 2°C and 4°C relative to pre-industrial levels (hereafter referred to as +2°K and +4°K, respectively).

These experiments are designed as 60-year equilibrium simulations rather than transient forecasts. While both the 2°C and 4°C warming levels are based on the RCP8.5 scenarios (van Vuuren et al., 2011), they employ constant external forcing fixed at the years 2040 and 2090, respectively. Consequently, the simulated years represent stable climate states at each specific warming level rather than chronological calendar years. Furthermore, by employing six patterns of future sea surface temperature and introducing perturbations to 15 initial conditions, a total of 90 ensemble simulations were conducted, allowing for probabilistic assessments. The d4PDF data set is used to probabilistically assess regional-scale characteristics of temperature and precipitation under future climate scenarios (e.g., Fujita et al., 2020; Ito et al., 2022; Sugimoto et al., 2024).

From the d4PDF data set, temperature and precipitation data over the region surrounding Japan under three different warming scenarios were used for proof-of-concept (PoC). Basic statistics, such as monthly average temperature and precipitation, along with probabilistic metrics derived from the 90 ensemble members (e.g., ensemble mean, standard deviation, and probability densities), were pre-calculated offline and formatted into tabular data (CSV files). Pre-calculating these metrics prevents the need for complex on-the-fly computations by the model or external modules during the query stage, thereby ensuring efficient data retrieval. To prepare for table retrieval using LLMs, the data were organized and stored as individual files by region, variable, scenario, and model, following the naming convention: [city/town name]_[variable]_[warming scenario]_[statistics].csv.

3. Methods

3.1. Model

3.1.1. Requirements and Overview

The fundamental requirements for an LLM designed to support climate services at the regional level are as follows:

1. Fundamental knowledge in climatology and global warming
2. Specific knowledge regarding adaptation measures implemented by local governments in Japan
3. Capability to utilize future climate projection data

For requirement 1., although models such as ClimateGPT (Thulke et al., 2024) and Arabic Mini-ClimateGPT (Mullappilly et al., 2023) have been proposed, this study adopts an approach based on extensibility, in which we collect data and generate training data sets ourselves, and perform fine-tuning using an existing base model. For requirement 2, since user needs vary depending on the context, we adopt a RAG approach (Gao et al., 2023; Lewis et al., 2020) to integrate external knowledge, enabling customizable responses. For requirement 3, we develop functionality that allows the model to interpret simulation data relevant to a given text query and generate responses based on that information.

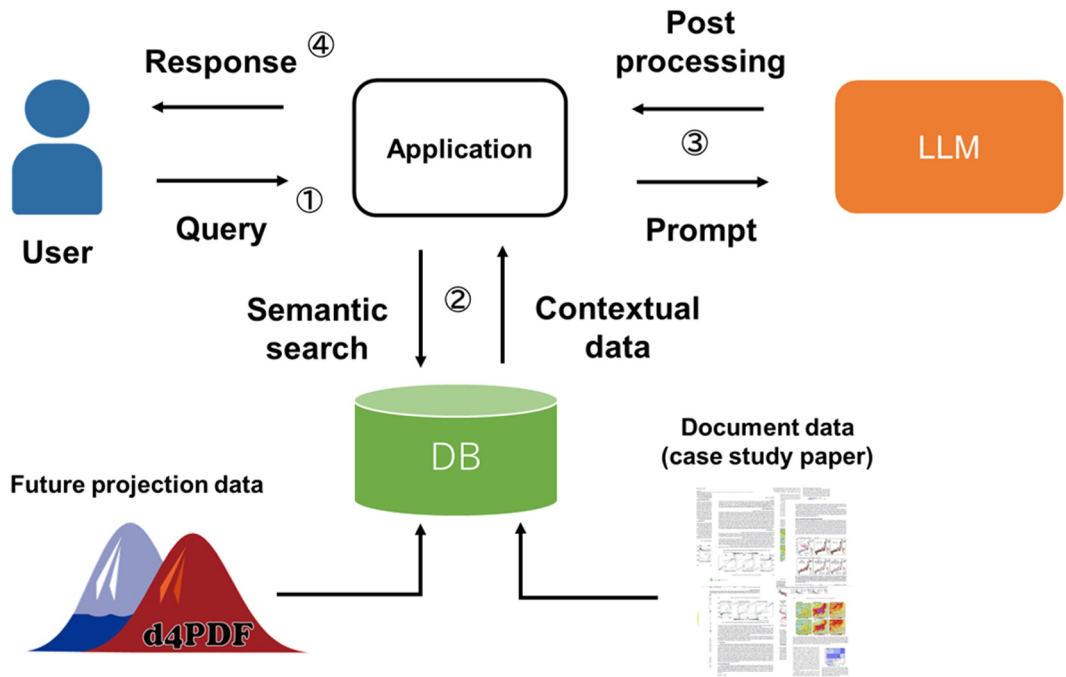


Figure 1. Overview of the proposed web-based application integrating an LLM with a vector database for climate adaptation support. d4PDF logo reproduced from the d4PDF project website: https://www.miroc-gcm.jp/d4PDF/.

Figure 1 illustrates the overall framework of a chatbot-type application that leverages an LLM to support climate change adaptation planning. The system is designed to generate context-aware responses to user queries regarding projected climate impacts at the local scale. The user inputs a natural language query, such as: "Please propose feasible adaptation measures considering the projected temperature and precipitation in [City Name] in [Year]." The application conducts a semantic search over a vector database (Vector DB) that contains encoded scientific documents (case study papers written in Japanese) and future projection data such as d4PDF. Document data are preprocessed and embedded in vector representations to facilitate efficient retrieval of relevant contextual information. The application constructs a prompt by combining the user query with the retrieved contextual data. This composite prompt is sent to the LLM, which generates a response incorporating domain-specific knowledge. For instance, it might reference projected temperature increases derived from d4PDF simulations or summarize findings from scientific papers. The generated response is optionally refined through post-processing before being presented to the user via the web interface. This architecture allows the system to deliver scientifically grounded and locally relevant adaptation strategies by integrating multimodal data sources with the reasoning capabilities of a general-purpose LLM.

3.1.2. Foundation Model

As the foundation model, this study adopts Llama 3.3 Swallow 70B Instruct v0.4 (https://swallow-llm.github.io/llama3.3-swallow.en.html), an open-source LLM developed by the Institute of Science Tokyo. This 70-billion-parameter model is built upon Meta's Llama 3.3 70B (https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct) through continued pretraining and fine-tuning. The pretraining process incorporates a Japanese web corpus (Swallow Corpus Version 2), Japanese and English Wikipedia articles, as well as mathematical and programming content. Additionally, fine-tuning with synthetic instruction data in Japanese enhances the model's instruction-following ability in Japanese while preserving its original English performance. For the original papers, please refer to Touvron et al. (2023) for LLaMA and Fujii et al. (2024) for LLaMA-Swallow. We transparently acknowledge that the foundation model's pre-training data, including Wikipedia and academic papers, likely overlap with the sources of our benchmark data sets (e.g., IPCC reports). Consequently, the model's performance may partially reflect memorization from pre-training. However, we ensured strict separation between the instruction-tuning data set and the benchmark data set, meaning no data leakage occurred during the fine-tuning

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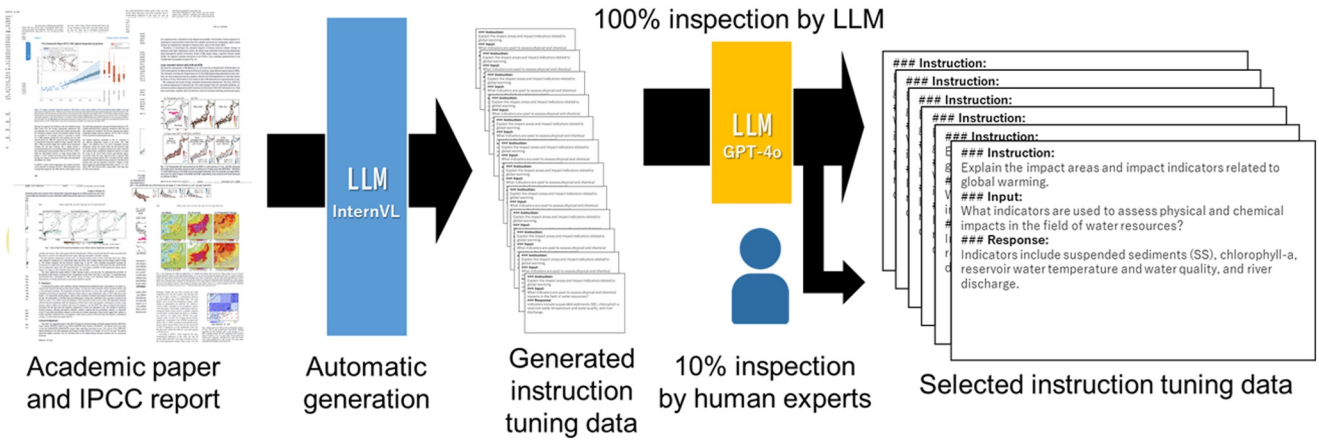


Figure 2. Generation of instruction-tuning data. Quality control is performed through a two-stage screening process: initial screening using an LLM and secondary screening by human experts.

phase. Our evaluation thus assesses the model's reasoning and synthesis capabilities rather than simple text memorization.

Leveraging this model's strong capabilities in Japanese, we performed domain-specific fine-tuning and developed a RAG system tailored to climatology and global warming. Although this model was selected with a focus on supporting climate change adaptation planning in Japan, the choice of foundation model is preferably aligned with the specific purpose and application context.

3.1.3. Data Preparation for Fine-Tuning and Performance Evaluation

To prepare for fine-tuning the foundation model and performance evaluation, instruction-tuning data were automatically generated from academic papers related to climatology and global warming, as well as from the Sixth Assessment Report of the IPCC. The workflow for generating and validating the instruction-tuning data is illustrated in Figure 2. Instruction-tuning refers to fine-tuning using supervised data consisting of question–answer pairs, optionally preceded by task instructions.

First, text was extracted from PDF files of the papers and reports. Then, a prompt. "Generate instruction, question, and answer sets from the provided academic content", was given to an LLM to generate instruction-tuning data. For PDF parsing, we used Docling, a Python package for document file analysis, and for data generation, we employed the InternVL2-5_78B-MPO model (https://github.com/OpenGVLab/InternVL), an open-source LLM. The generated question-answer (QA) sets comprise two formats, multiple-choice QA and short-answer QA, and are provided in both English and Japanese (the latter translated from English).

The automatically generated data included instances of character corruption and formatting errors. To address this, we employed InternVL to detect and remove such deficiencies automatically. In addition, a 10% subset of the filtered data was manually reviewed by domain experts to verify the scientific accuracy of the content. As a result of the validation process, 190,120 out of the initially generated 191,590 QA pairs passed both automated and manual validation and were retained for instruction-tuning and evaluation. To clarify, while all retained pairs successfully passed the automated filtering, approximately 10% of them also passed the subsequent manual review, whereas the remaining 90% were validated by the automated filtering alone.

3.1.4. Fine Tuning

Fine-tuning of the base model was conducted using the generated instruction-tuning data set. Through this process, the model's parameters were updated to produce correct answers in response to given inputs, thereby enabling the acquisition of knowledge related to global warming. There are two main approaches to fine-tuning: full fine-tuning, which updates all model parameters, and a parameter-efficient method that updates only a subset of parameters reduced through low-rank approximation (LoRA; Hu et al., 2021). In this study, we adopted the

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latter approach, a parameter-efficient fine-tuning method that applies low-rank matrix decomposition to weight updates, thereby reducing computational cost while preserving the base model's capabilities.

The data set was randomly split into training, validation, and test sets in a ratio of 8:1:1. In Experiment 1, the learning rate was set to 1.0×10^{-4}, and in Experiment 2, to 2.0×10^{-5}. The highest validation accuracy was achieved at 19,000 iterations in Experiment 1 and 36,500 iterations in Experiment 2. The maximum test accuracies were 0.8142 and 0.8112, respectively. The model from Experiment 1 was used in subsequent evaluations.

In addition, to improve the model's ability to interpret simulation data and perform scientific reasoning, we also conducted fine-tuning using TableLlama (Zhang et al., 2023), specialized for instruction-tuning with tabular data, and ScienceQA (Lu et al., 2022), a benchmark data set for scientific question answering.

3.1.5. Retrieve Augmented Generation (RAG) for Text Data

To improve the output quality of the developed LLM by providing the model with specific external knowledge tailored to the user's objectives, an RAG framework was implemented. RAG enhances response quality by retrieving relevant information from an external knowledge base in response to a user query, and incorporating that information into the language model's input. This allows the system to generate accurate outputs grounded in up-to-date knowledge, even if such information is not encoded in the model parameters.

In our implementation, we collected 227 peer-reviewed papers from A-PLAT, specifically focused on climate change adaptation measures unique to Japan. Each document was converted to plain text, segmented into paragraph-level units, and embedded in a high-dimensional vector space using a sentence embedding model. These vectors were stored in a vector database, enabling semantic similarity search across the document corpus.

At inference time, the user query is embedded and compared to entries in the vector database to retrieve semantically relevant document chunks. These retrieved texts are appended to the prompt and provided to the LLM for final response generation. This approach enables the model to dynamically reference external sources without additional training, thereby supporting accurate context-aware responses to a broad range of climate-related queries.

3.1.6. Retrieve Augmented Generation (RAG) for Future Projection Data

In this study, we utilize climate change projection data derived from climate models to generate user-specific responses that support the formulation of more concrete adaptation measures. To achieve this, we developed a method to extract numerical climate projection data based on user text queries (Figure 3). The proposed method consists of four modules: Query Structuring, Search Query Generation, CSV Retrieval, and Answer Generation. The Query Structuring module takes a user-input text query and, using an LLM, converts it into a structured format to produce analysis conditions suitable for identifying relevant files. The Search Query Generation module then generates the appropriate file name(s) by matching the analysis conditions to a given list of file names. In the CSV Retrieval module, the system accesses the climate projection database to retrieve the tabular data corresponding to the identified file name(s). Finally, the Answer Generation module extracts and utilizes the relevant numerical values from the retrieved table to construct a response to the user's query. Since the d4PDF data set consists of a 90-member ensemble, dynamic computation during the query stage would introduce significant overhead. Therefore, as described in Section 2.2, probabilistic metrics are precalculated and stored in the CSV files.

To address the methodological formulation, we formalize the CSV-based RAG through four computational steps. First, in Query Structuring, let L, V, T, W, and M be the sets of locations, climate variables, time periods, warming scenarios, and statistical metrics, respectively. We define the multi-dimensional coordinate space of the future projection data as the Cartesian product $S = L \times V \times T \times W \times M$. An LLM parameterized by θ maps a natural-language query q into a structured coordinate vector $c \in S$. Subsequently, in Search Query Generation and CSV Retrieval, a retrieval mapping function $g(c)$ identifies and retrieves the corresponding tabular data set Dc from the database based on the extracted coordinates. Thus, the retrieval process is defined as $Dc = g(c)$. We mathematically formalize an uncertainty aggregation mechanism, $\mathrm{Agg}(Dc, m)$, not as a dynamic calculator, but as an extraction function. It retrieves the pre-calculated, actionable statistical indicators (e.g., mean μ, standard deviation σ, or specific percentiles) corresponding to metric m directly from the retrieved tabular data set Dc. Finally, in Answer Generation, the aggregated statistical information and the original query q are concatenated

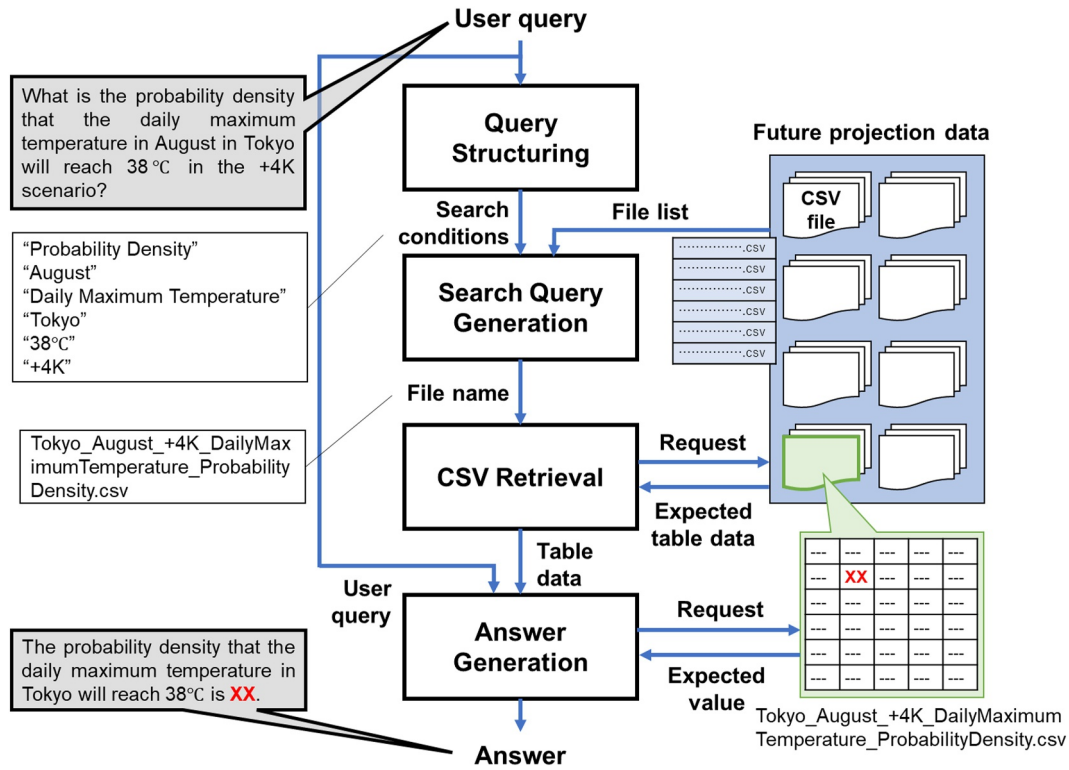


Figure 3. Utilization of climate projection data in an LLM. Relevant files are retrieved based on a user's text query, the corresponding data are extracted, and a response is generated.

and fed into the LLM to generate the final response a, formulated as $a = \mathrm{LLM}_\theta(q \oplus \mathrm{Agg}(Dc, m))$, where $\oplus$ denotes the concatenation in the prompt.

This feature enables users to extract probabilistic information regarding specific events from the d4PDF ensemble data sets. For example, in response to a user query such as "What is the probability density that the daily maximum temperature in August in Tokyo will reach 38°C in the +4K scenario?", the LLM alone does not possess sufficient knowledge to provide a precise answer. In such cases, CSV-based RAG is utilized. By structuring the user query, a set of analysis conditions is extracted: "Probability Density," "August," "Daily Maximum Temperature," "Tokyo," "38°C," and "+4K." Using these conditions to search the list of file names, a candidate file such as "Tokyo_August_+4K_DailyMaximumTemperature_ProbabilityDensity.csv" is identified. From the tabular data within this file, the relevant numerical value corresponding to the user query is retrieved, enabling the system to generate a fact-based response such as: The probability density that the daily maximum temperature in Tokyo will reach 38°C is XX." In the implementation of this module, we conducted preliminary testing across 27 patterns involving various queries and CSV files. The results confirmed that the numerical information was correctly extracted with an accuracy of 96.3%. When ambiguous queries lacking specific indicators or time horizons are input into the prompt, the model may output erroneous values. Therefore, it is necessary to either provide appropriate prompts based on a clear understanding of the registered CSV files or implement additional error-handling procedures. From a usability perspective, the system should incorporate default prompt configurations that automatically supplement missing parameters, such as warming scenarios or temporal resolutions, alongside a guided interface informing users of the available data range to effectively mitigate the risk of query failure for less-experienced practitioners.

3.1.7. Web Application

We implemented an interactive web application that integrates both text-based RAG and CSV-based RAG into the fine-tuned climate-specialized LLM. The base of the application was developed using LangChain (https://www.langchain.com), a framework designed for building applications with LLMs. The user interface was then

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Table 1
Model Performance Comparison on the Climatology-Specific Benchmark

	English task										
	Agriculture, forestry and fishery (362)	Water environment and water resources	Natural ecosystems (343)	Disasters and coastal area (94)	Health (228)	Industry and economy (30)	Daily life and urban life (43)	Others (278)	Physical science basis (793)	Impacts, adaptation, vulnerability (1244)	Mitigation (809)
Ours (with RAG)	**0.716** (0.904)	**0.769** (0.923)	**0.757** (0.898)	**0.722** (0.904)	**0.742** (0.912)	0.697 (0.939)	**0.654** (0.904)	**0.750** (0.898)	**0.730** (0.902)	**0.894** (0.939)	**0.885** (0.943)
Swallow 70B	0.695	0.662	0.664	0.67	0.714	0.606	**0.654**	0.622	**0.73**	0.853	0.838
GPT-4o	0.678	0.677	0.683	0.687	0.678	**0.727**	0.615	0.648	0.72	0.849	0.831
	Japanese task										
	Agriculture, forestry and fishery (255)	Water environment and water resources	Natural ecosystems (305)	Disasters and coastal areas (69)	Health (180)	Industry and economy (37)	Daily life and urban life (28)	Others (223)	Physical science basis (695)	Impacts, adaptation, vulnerability (1056)	Mitigation (657)
	0.673 (0.665)	**0.713** (0.701)	**0.698** (0.702)	**0.688** (0.645)	**0.728** (0.701)	**0.723** (0.681)	**0.718** (0.667)	**0.711** (0.695)	**0.815** (0.806)	**0.881** (0.881)	**0.865** (0.853)
	0.579	0.655	0.58	0.656	0.572	0.575	0.539	0.617	0.732	0.843	0.797
	0.598	0.632	0.621	0.613	0.634	0.532	**0.718**	0.605	0.745	0.84	0.772

Note. "Ours" denotes the fine-tuned model without text RAG, and "with RAG" denotes the same model with text RAG (values shown in parentheses). Swallow 70B refers to Llama 3.3 Swallow 70B Instruct v0.4, and GPT-4o refers to gpt-4o-2024-08-06. The highest score in each category is highlighted in bold.

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implemented using Chainlit (https://chainlit.io), a framework for developing conversational AI systems. Additionally, a web search function was incorporated into the application, enabling it to retrieve up-to-date information from web pages when necessary. The web search function is implemented to be toggled on or off during use. For both the benchmark performance evaluation (Section 4.1) and the case study (Section 4.2), the experiments were conducted with the web search function disabled.

3.2. Evaluation

3.2.1. Climatology-Specific Benchmark

In this study, we evaluated the model's ability to generate responses related to climatology and global warming using a climate-specific benchmark data set. As a benchmark, we used the multiple-choice QA subset (excluding short-answer QA) of the 10% test split that was not used for model development. The benchmark set consists of 4,275 QAs in English and 3,560 QAs in Japanese, for a total of 7,835 QAs. As with the instruction-tuning data, the questions are categorized into the 11 domains including "Others" described in 2.1. The number of QA pairs varied substantially by category: in both languages, the largest category, "Impacts, adaptation, and vulnerability," contained roughly 40 times more pairs than the smallest categories, "Industry and economy" and "Daily life and urban life", as shown in Table 1.

An example of the benchmark questions is shown in Table S1. Our benchmark consists of passage-grounded multiple-choice QA items: a "Content" passage is paired with a templated "Prompt" that presents the "Question" and "Answer" options to the model. The model's prediction is scored by matching it to the gold "Answer", while the "Explanation" provides a human-readable justification aligned with the passage to verify that the item is answerable solely from the provided context.

3.2.2. General-Purpose Benchmark

We evaluated the model's performance on general tasks using benchmark data sets. It is important to assess whether the fine-tuning process has led to any degradation in the model's general capabilities. In this study, we used the swallow-evaluation as the benchmark data sets. Swallow-evaluation is a benchmark suite designed for evaluating the Llama Swallow model (https://github.com/swallow-llm/swallow-evaluation). It consists of three components: MT-Bench JA, Japanese tasks, and English tasks. We also employed MMLU-Pro (Wang et al., 2024) to rigorously evaluate the model's advanced reasoning capabilities and specialized knowledge across diverse domains.

MT-Bench JA is a benchmark for multi-turn conversational abilities in Japanese, with a particular focus on instruction comprehension, consistency and accuracy of information generation, conversational coherence, and cultural and linguistic naturalness across eight domains: Coding, Extraction, Humanities, Math, Reasoning, Roleplay, STEM, and Writing.

To further assess specialized capabilities, we utilized a suite of Japanese and English tasks. The Japanese benchmarks evaluate common sense reasoning (JCom; Kurihara et al., 2022), multi-hop question answering (JEMHopQA; Ishii et al., 2024, NIILC; Sekine, 2003), reading comprehension (JSQuAD; Kurihara et al., 2022), summarization (XL-Sum; Hansen et al., 2021), and mathematical reasoning (MGSM; Shi et al., 2023). It also includes translation (WMT20; Barrault et al., 2020), general knowledge (JMMLU; Yi et al., 2024), and code generation (JHumanEval; Sato et al., 2024).

Similarly, the English benchmarks measure capabilities in open-domain and science question answering (OpenBookQA; Mihaylov et al., 2018, TriviaQA; Joshi et al., 2017, SQuAD2; Rajpurkar et al., 2018), linguistic reasoning (HellaSwag; Zellers et al., 2019, XWINO; Tikhonov & Ryabinin, 2021), general knowledge (MMLU; Hendrycks et al., 2021), and complex problem-solving in mathematics (GSM8K; Cobbe et al., 2021, MATH; Hendrycks et al., 2021), logic (BBH; Suzgun et al., 2023), and programming (HumanEval; Chen et al., 2021).

The evaluation metrics vary depending on the task type. For MT-Bench JA, we employed an LLM-as-a-judge approach, where a strong language model (GPT-4) scores the model's responses on a scale of 1–10 based on predefined rubrics. For the Japanese and English tasks, performance is primarily measured using Exact Match (EM) or F1-score for question-answering and classification tasks, and Accuracy for multiple-choice problems (e.g., MMLU, JMMLU). Coding tasks (HumanEval, JHumanEval) are evaluated using the pass@k metric, which

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Table 2
Model Performance Comparison on General-Purpose Benchmarks

	Japanese MT-Bench							
Model	Coding	Extraction	Humanities	Math	Reasoning	Roleplay	STEM	Writing
Ours	0.543	0.727	0.644	0.498	0.505	**0.881**	0.497	0.74
Swallow 70B	0.705	0.82	0.87	0.73	0.623	0.811	0.781	0.832
GPT-4o	**0.855**	**0.926**	**0.88**	**0.872**	**0.706**	0.862	**0.838**	**0.849**

Japanese task									
Jcom	JEMHopQA	NIILC	JSQuAD	XL-Sum	MGSM	WMT20 (E-J)	WMT20 (J-E)	JMMLU	JHumanEval
0.953	0.623	0.672	**0.909**	0.137	0.76	0.308	0.25	0.694	–
0.981	0.618	0.662	0.907	0.162	0.812	**0.319**	**0.261**	0.707	0.7
0.982	**0.731**	**0.709**	0.889	**0.17**	**0.864**	0.314	0.254	**0.797**	**0.752**

English task									
OpenBookQA	TriviaQA	HellaSwag	SQuAD2	XWINO	MMLU	GSM8K	MATH	BBH	HumanEval
0.43	0.809	–	0.586	**0.915**	0.795	0.873	**0.649**	0.806	–
0.448	**0.817**	**0.686**	**0.654**	0.912	**0.803**	**0.908**	0.566	**0.812**	**0.75**
–	–	–	–	–	–	–	–	–	–

MMLU-Pro													
Biology	Chemistry	Physics	Engineering	Math	CS	Health	Economics	Business	Law	History	Philosophy	Psychology	Other
0.799	0.509	0.547	0.518	0.565	0.622	0.663	0.706	0.578	0.384	0.588	0.609	0.749	0.629
0.798	0.551	0.54	0.44	0.591	0.632	0.644	0.709	0.597	0.386	0.585	0.557	0.712	0.618
0.893	**0.727**	**0.751**	**0.553**	**0.794**	**0.783**	**0.76**	**0.816**	**0.801**	**0.59**	**0.732**	**0.703**	**0.827**	**0.796**

Note. Results for GPT-4o on the English task are not publicly available and are therefore omitted here. The highest score in each category is highlighted in bold.

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measures the probability that at least one of the top-k generated code samples passes all unit tests. MMLU-Pro specifically utilizes a 10-way multiple-choice format to rigorously test reasoning accuracy. For more detailed methodologies of each benchmark, please refer to the respective cited literature.

4. Results

4.1. Performance Evaluation on Benchmark

4.1.1. Climatology-Specific Performance

The developed climatology-specific LLM without text RAG demonstrated consistently high performance across both English and Japanese tasks (Table 1). Compared to Swallow 70B and GPT-4o, it achieved the highest scores in the majority of categories, including "Natural ecosystems," "Disasters and coastal area," and "Health." Notably, for Japanese-language tasks, the model outperformed the baselines in every category, indicating that domain-specific fine-tuning significantly improved its ability to understand climate-related content in both languages. These results highlight the effectiveness of our model in handling technical terminology and structured knowledge specific to climatology, even without relying on external retrieval mechanisms. It is important to note the conditions of these comparisons. Because our model and Swallow 70B share the identical pre-training corpus, the comparison between them is strictly fair, directly demonstrating the effectiveness of the domain-specific fine-tuning. In contrast, since the pre-training data of proprietary models such as GPT-4o are undisclosed, a strictly fair evaluation regarding pre-training overlap is impossible. Therefore, the comparison with GPT-4o should be interpreted as reference information rather than an absolute benchmark.

Despite its strengths, the developed model without text RAG showed relatively lower performance in certain categories when compared to state-of-the-art general-purpose models. e.g., in the English task, GPT-4o slightly outperformed our model in "Industry and economy," although the differences were marginal (e.g., 0.727 for GPT-4o vs. 0.697 for our model). While our model maintained a significant lead in technical fields like "Disasters and coastal areas", this performance gap was less pronounced in categories requiring broader linguistic flexibility. These results indicate that the advantages of domain-specific fine-tuning vary depending on the socioeconomic or general nature of the category.

For reference, we also evaluated the performance of the same model with RAG integrated. The results show further improvements in nearly all categories, particularly in knowledge-intensive domains such as "Impacts, adaptation and vulnerability" and "Mitigation" in the English task. The RAG-enhanced model achieved the highest scores overall, for instance, 0.939 with RAG versus 0.894 without RAG for "Impacts, adaptation and vulnerability." Although these results are provided as supplementary data, they suggest that incorporating domain-specific external documents through retrieval-augmented generation can further enhance the contextual accuracy and completeness of generated responses.

4.1.2. General Performance

We evaluated the general-purpose performance of the developed climate-specialized LLM by comparing it with Swallow 70B and GPT-4o across Japanese and English benchmarks (Table 2). As expected, the proposed model did not outperform the large-scale general-purpose models, especially GPT-4o, in most tasks. Furthermore, we observed noticeable performance degradation in reasoning-heavy categories. For instance, in the Japanese MT-Bench, scores for STEM, Math, and Humanities fell substantially compared to the base Swallow 70B model (e.g., STEM fell from 0.781 to 0.497, and Math fell from 0.730 to 0.498). While the model maintained competitive scores in conversational tasks such as "Roleplay" (0.881) and "Writing" (0.740), the drop in other areas highlights a clear trade-off between acquiring specialized climate knowledge and maintaining general reasoning capabilities.

In the Japanese task benchmarks (e.g., JCom, JEMHopQA, JSQuAD), the proposed model achieved generally strong performance. It performed comparably with Swallow 70B in QA tasks such as JEMHopQA (0.623 vs. 0.618), JSQuAD (0.909 vs. 0.907), and NIILC (0.672 vs. 0.662). In English benchmarks, our model's performance was somewhat lower than the baselines, particularly in standard answering tasks (SQuAD: 0.654 for Swallow 70B vs. 0.586 for ours). However, it still achieved reasonable scores in math and common sense such as MATH (0.649) and XWINO (0.915). These numerical results confirm the model's baseline capabilities across both general-purpose and climate-specific tasks.

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Prompt 1
Based on the RCP8.5 scenario under a +4℃ warming condition, provide the ensemble mean, +σ, and −σ values of the daily maximum temperature in July for Kumagaya City. How much do these values increase compared to the average daily maximum temperature in July 2020 in Kumagaya City (27.6℃)? Additionally, what potential risks could arise from these temperature increases?

Response 1
Under the RCP8.5 scenario with a +4℃ warming condition, the ensemble mean of the daily maximum temperature in July for Kumagaya City is 33.0℃, with a +σ value of 35.1℃ and a −σ value of 30.9℃.
Compared to the average daily maximum temperature in July 2020 in Kumagaya City (27.6℃), these values represent increases of 5.4℃, 7.5℃, and 3.3℃, respectively.
Such extreme temperature increases may lead to heightened health risks, including a greater incidence of heat-related illnesses such as heatstroke.

Prompt 2
Using the projected increases in daily maximum temperature from July 2020—5.4℃ for the average scenario, 7.5℃ for the pessimistic scenario, and 3.3℃ for the optimistic scenario—calculate the expected increase in the number of heatstroke patients for each level: average, pessimistic, and optimistic.
Taking into account the increase in heatstroke patients, how many additional green curtains and rest spots need to be installed as part of the current heat mitigation measures? Please first present the relationship equations used in Kumagaya City to determine the required number of additional green curtains and rest spots, and then show the required number of additional installations for each measure together with the calculation process.

Response 2
The increase in the number of heatstroke patients can be calculated as:
20 × increase in daily maximum temperature for each level
For the average scenario: 20 × 5.4 = 108 additional patients
For the pessimistic scenario: 20 × 7.5 = 150 additional patients
For the optimistic scenario: 20 × 3.3 = 66 additional patients

Since one additional green curtain is required for every 50 additional heatstroke patients, the required numbers are as follows:
For the average scenario: 108 / 50 = 2.16, so 2 additional green curtains are required.
For the pessimistic scenario: 150 / 50 = 3, so 3 additional green curtains are required.
For the optimistic scenario: 66 / 50 = 1.32, so 1 additional green curtain is required.

Since one additional rest spot is required for every 10 additional heatstroke patients, the required numbers are as follows:
For the average scenario: 108 / 10 = 10.8, so 11 additional rest spots are required.
For the pessimistic scenario: 150 / 10 = 15, so 15 additional rest spots are required.
For the optimistic scenario: 66 / 10 = 6.6, so 7 additional rest spots are required.

Figure 4. Case study on the planning of heat adaptation measures in Kumagaya City, Saitama Prefecture. Multiple strategies are proposed based on statistical information derived from climate change projection data.

4.2. Case Study for Proof of Concept

A case study was conducted to demonstrate how future climate projection data can be utilized to quantitatively and concretely update existing heat-mitigation measures. The target area was Kumagaya City in Saitama Prefecture, Japan, which is known as one of the regions facing critical heat-related challenges, as summer maximum temperatures often exceed 40°C due to urbanization and topographic factors. In this case study, multiple heat-mitigation strategies were developed based on probabilistic climate change projection data under the 4°C warming scenario, designed as a 60-year equilibrium simulation using the external forcing of 2090 under the RCP 8.5. While these projections reflect a long-term state, they are utilized here for short-term measures, such as green curtains and cooling spots, to serve as a PoC. Given that the impacts of warming are already manifesting in

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Prompt 3
Three experts (a researcher, a Kumagaya City official, and a consultant) are asked to compare the following three heat mitigation options and develop a single, realistic, and effective integrated plan.

Options
(1) +5.4℃ : 2 green curtains, 11 rest areas
(2) +7.5℃ : 3 green curtains, 15 rest areas
(3) +3.3℃ : 1 green curtain, 7 rest areas

Roles
- Researcher: evaluates scientific validity and effectiveness
- Kumagaya City official: evaluates feasibility, cost, and public accountability
- Consultant: proposes a practical integrated plan

Rules
- The discussion must include at least two rounds, and each expert should revise their opinion based on others' input
- The final output must be a single integrated plan, not a majority decision
- Elements from different options may be combined if necessary

Output format
[Round 1] Initial opinions of each expert
[Round 2] Revised opinions based on discussion
[Final] Integrated plan (number of green curtains and rest areas) with justification (scientific validity, feasibility, and overall appropriateness)

Response 3
[Round 1]
Researcher: From a scientific perspective, the option corresponding to a +7.5℃ increase provides the highest level of protection, as it maximizes both green curtains and rest areas.
Kumagaya City official: Considering budget constraints, the +7.5℃ option may be financially challenging. It is also important to clearly justify the necessity and effectiveness of the measures to the public.
Consultant: Taking into account both the scientific evidence and financial constraints, a balanced approach is recommended. The +5.4℃ option (2 green curtains and 11 rest areas) provides effective protection while maintaining financial feasibility.

[Round 2]
Researcher: Given the financial constraints, the +5.4℃ option appears more realistic. However, additional measures should be considered if higher levels of protection become necessary.
Kumagaya City official: The consultant's proposal is financially feasible and easier to justify to the public. However, continuous monitoring and evaluation will be necessary to address future temperature increases.
Consultant: Integrating both perspectives, the +5.4℃ option should be adopted as a baseline, with flexibility to scale up measures depending on future conditions.

[Final]
Integrated plan: 3 green curtains and 11 rest areas.
Justification: This plan balances scientific validity, feasibility, cost-effectiveness, and public accountability, making it the most appropriate option under the given conditions.

Figure 4. (Continued)

regions like Kumagaya City, this approach intentionally leverages advanced ensemble projections to proactively evaluate localized adaptation strategies. It should be noted that this case does not aim to substitute human decision-making. Rather, the objective is to demonstrate how the system can be utilized in the early stages of

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planning to rapidly generate a data-informed baseline that integrates both scientific projections and local constraints.

Figure 4 presents the prompts and responses used in the case study. The process was structured into three sequential prompts, covering the extraction of climate projection data, the development of multiple countermeasures based on probabilistic information, and the integration of these into a final plan. In Prompt 1, the LLM was asked how much the statistical metrics of daily maximum temperature in July under the RCP8.5 scenario (ensemble mean, $+\sigma$, and $-\sigma$) would increase compared with the observed July 2020 value of 27.6°C. The model correctly extracted values of 33.0°C, 35.1°C, and 27.6°C from the tabular ensemble data, corresponding to increases of 5.4°C, 7.5°C, and 3.3°C, respectively (lines 9–13). In response to the query regarding potential risks associated with such temperature increases, the model identified the elevated risk of heatstroke and other health-related impacts (lines 14–15).

In Prompt 2, the temperature-increase values extracted in Prompt 1 were applied to refine existing heat-mitigation measures in Kumagaya City. These include the installation of green curtains, in which plants are grown to cover windows and block sunlight, and the creation of rest spots, which serve as cooling shelters where people can temporarily escape the heat. By using the ensemble mean, $+\sigma$, and $-\sigma$ values, three levels of countermeasures were generated, corresponding to average, pessimistic, and optimistic scenarios. Crucially, rather than relying on explicitly prescribed calculation rules within the prompt, the model utilized the RAG framework to dynamically retrieve context-appropriate logic and local constraints from existing municipal adaptation guidelines. To provide interpretability and demonstrate the RAG capabilities transparently, Prompt 2 was designed to explicitly require the model to first present these retrieved local constraints before applying them. In its response, the model successfully presented the retrieved formulas—specifically, that the increase in heatstroke patients is calculated as 20 times the temperature increase, and that one additional green curtain and rest area are required for every 50 and 10 additional patients, respectively. Based on the projected temperature increases and these presented parameters, the model showed the step-by-step calculation process and estimated additional heatstroke patients at 108 in the average case, 150 in the pessimistic case, and 66 in the optimistic case, while explicitly displaying the calculation procedure in accordance with the retrieved local constraints (lines 28–32). Furthermore, the system translated these health-risk estimates into concrete infrastructure requirements, determining that 2 green curtains would be needed for the mean scenario, with 3 required for the pessimistic case and 1 sufficient for the optimistic case. Similarly, it estimated that 11, 15, and 7 additional rest areas would be necessary for the respective scenarios. These values are correctly calculated based on the Kumagaya City guidelines retrieved via RAG, which prescribe the installation of one additional green curtain for every 50-person increase in heatstroke patients and one additional rest area for every 10-person increase. This step illustrates how the system moves beyond simple data retrieval and scripted calculations, effectively integrating probabilistic climate data with region-specific knowledge to produce tailored, actionable baselines for local planners.

In Prompt 3, the multiple levels of countermeasures generated in Prompt 2 were consolidated into a final plan. The outputs in Prompt 2 ranged between 1–3 green curtains and 7–15 rest spots. While increasing these numbers could reduce heatstroke cases, excessive measures would also result in unnecessary costs. To balance these trade-offs, three expert roles, namely scientist, consultant, and city official, were introduced to conduct a structured deliberation from different perspectives. The scientist provided data-driven analysis, the consultant emphasized effectiveness over cost, and the city official assessed practical feasibility with a focus on financial constraints (lines 55–58).

The LLM-generated response produced a collaborative discussion among a researcher, a consultant, and a city official, resulting in an integrated heat-mitigation strategy for Kumagaya City. The dialog progressed as the researcher presented scientific evidence for temperature rise scenarios, the consultant emphasized effective measures and the importance of public and business engagement, and the city official evaluated feasibility and cost considerations. Ultimately, the discussion converged on a phased implementation beginning with measures for a 5.4°C rise, incorporating citizen participation to enhance sustainability, thereby balancing scientific validity, policy effectiveness, cost management, and long-term resilience. It is important to emphasize that this role-playing simulation is not intended to act as a proxy for actual stakeholder engagement. Instead, it systematically maps out potential trade-offs from divergent priorities to consolidate multiple scenarios into a structured preliminary baseline. This process demonstrates the LLM's ability to simulate consensus-building among stakeholders with divergent priorities.

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5. Discussion and Conclusions

This study aimed to develop a climatology-specific LLM capable of utilizing future projection data, and to evaluate its effectiveness in supporting the planning of climate change adaptation measures. The developed model demonstrated competitive performance compared to general-purpose models on both English and Japanese climatology-specific benchmarks, with particularly notable improvements in the domains of "mitigation" and "impacts, adaptation, and vulnerability." Through the integration of CSV-based RAG, the model was able to access quantitative temperature and precipitation data, enabling the generation of actionable data-driven recommendations. Case studies simulating heat adaptation planning for Kumagaya City demonstrated the technical feasibility of the model in integrating probabilistic projections with predefined adaptation measures and calculation logic to generate structured plans. While further evaluation against real-world planning criteria and stakeholder feedback is necessary, these initial results suggest that LLMs have the potential to serve not only as text generators but also as promising preliminary support tools for decision-making in climate-related contexts. It must be emphasized that it is entirely unrealistic for complete non-experts to independently formulate viable climate adaptation plans. Effective planning inherently requires generating multiple countermeasures from diverse perspectives and carefully selecting the most practical and effective options. The framework proposed in this study is not intended to replace human expertise, but rather to amplify the capacity of municipalities where expert resources are scarce. By rapidly generating diverse, data-informed baselines, the LLM assists these limited practitioners in the complex selection process. As a proof-of-concept, our results successfully demonstrate the technical feasibility of this LLM-assisted workflow, paving the way for future operational testing.

Recent years have seen the development of LLMs specialized in geoscientific domains, such as ClimateBERT (Webersinke et al., 2021), ClimateBERT-NetZero (Schimanski et al., 2023), ClimateGPT (Thulke et al., 2024), OceanGPT (Bi et al., 2023), and K2 (Deng et al., 2024). These models have shown strengths in understanding technical terminology and summarizing scientific literature; however, they do not support direct integration of quantitative projection data. In contrast, our model combines domain-specific fine-tuning with the ability to retrieve and reference structured external data (in CSV format) through RAG, representing a novel methodological contribution. Furthermore, this study incorporated case studies closely aligned with real-world adaptation practices of local governments and businesses, allowing us to evaluate the practical applicability of the model. To our knowledge, this is the first study to systematically demonstrate the utility of the climatology-specific LLM not only in terms of general language capabilities but also in supporting concrete policy development and climate risk assessment.

Performance evaluation across both climate-specific and general-purpose benchmarks reveals a critical balance between specialized knowledge acquisition and the retention of foundational linguistic abilities. Our results indicate a known trade-off in LLM development: acquiring specialized climate knowledge results in partial catastrophic forgetting, leading to a degradation in general reasoning tasks such as STEM and Math. Nevertheless, the model maintains sufficient foundational conversation capabilities for tasks beyond climate science, making it suitable for integration into general-purpose workflows. We attribute this retention of basic conversational abilities in part to the use of LoRA, which updates only a subset of parameters, thereby helping to preserve the foundation model's broad language understanding.

However, we observed that general-purpose models like GPT-4o maintain a slight advantage in categories involving broader societal contexts or complex socioeconomic patterns. This suggests that while specialized fine-tuning excels in technical domains, additional strategies may be needed to further improve performance in everyday or socioeconomically diverse contexts. In practice, acquiring new specialized knowledge and retaining prior competencies are inherently in tension, reflecting a known trade-off in LLM development (e.g., Haque, 2025). Given these considerations, our framework demonstrates a reasonable compromise, successfully bridging high-level technical accuracy with robust general-purpose utility. Nevertheless, ensuring the reliability of these outputs remains a priority as the system transitions from a research prototype to a practical tool.

Despite its contributions, this study has several limitations and challenges. First, there remains a risk that the LLM may produce incorrect reasoning or hallucinated content. Automatically generated adaptation measures should not be implemented without expert review, and future work should address the need for built-in validation and fact-checking mechanisms. While this study performed fine-tuning using a data set of 338 peer-reviewed papers and IPCC reports supervised by climate adaptation experts, incorporating a broader range of high-quality data including via RAG remains essential for further suppressing hallucinations. It is important to recognize that LLMs

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serve as the preliminary support tools designed to complement rather than substitute real participatory and governance-based decision-making processes. The role-playing simulation presented in our case study is intended solely to provide a structured, data-informed baseline to facilitate subsequent human deliberation and not to act as a proxy for actual stakeholder engagement. To rigorously evaluate this capability, future research must benchmark the AI-generated outcomes against independent, real-world stakeholder processes, such as blind comparisons with human municipal practitioners. Second, the model currently relies on pre-registered climate projection data sets, which restricts its flexibility. To address this, several strategies can be considered: conducting user needs assessments to identify relevant regions, variables, and bias correction methods for data registration; allowing users to upload custom data sets for tailored model behavior; or enabling the model to automatically generate keys to query external databases such as d4PDF. These enhancements will be essential to ensure that the model remains responsive to diverse and evolving climate information needs. Third, the reliance on established adaptation plans may bias the model toward incremental solutions rather than transformative strategies. This is particularly critical under high-emission scenarios such as RCP8.5, where historical precedents may no longer suffice for unprecedented climatic shifts. Future research should explore integrating speculative climate scenarios or multi-sectoral foresight modeling to encourage novel long-term adaptation pathways. Balancing expert-verified reliability with the creative exploration of transformative responses remains a vital challenge for enhancing the system's strategic utility. Finally, regarding our benchmark evaluation, we must explicitly note potential limitations arising from our evaluation design and data. Because both the instruction-tuning data and the benchmark data sets were generated using the same automated process (InternVL), they share identical underlying data distributions. Although our evaluation relied exclusively on multiple-choice questions, which mitigates the risk of the model simply mimicking free-text output styles, the model may still have adapted to specific linguistic patterns or question-formatting biases inherent to the generator. To isolate true domain generalization, future research must rigorously evaluate the model using completely independent, human-curated data sets. Furthermore, we must acknowledge the potential limitation of pre-training data overlap. While there is strictly no overlap between our instruction-tuning data and the benchmark data, the foundation model's extensive pre-training corpus likely contains climate literature. Thus, we have refrained from making definitive claims of outperforming proprietary models and instead framed such comparisons as references.

By lowering the barriers to expert-level climate analysis, the proposed framework has the potential to significantly enhance the capacity of local governments to respond proactively to climate risks. In practice, local decision-makers can use this system in the early stages of adaptation planning to generate data-informed baselines for specific questions, such as quantifying the required heat-mitigation infrastructure under specific warming scenarios. A significant advantage of utilizing an LLM is its capability to rapidly iterate and simulate numerous such scenarios. By exploring a wide array of simulated case studies under varying parameters and local constraints, human decision-makers can ultimately leverage these diverse data-informed baselines to make more robust and comprehensive final choices. However, to make the outputs truly actionable, users must integrate local context data, such as specific vulnerability, exposure, adaptive capacity, institutional constraints, and financial resources, through carefully structured prompts. Reliable and context-appropriate results require prompts that explicitly define step-by-step calculation logic, local socio-economic constraints, and diverse stakeholder roles to evaluate trade-offs. The integration of advanced AI with such context-rich prompts and climate projection data enables the generation of tailored adaptation strategies that reflect regional and sector-specific vulnerabilities and needs. This, in turn, fosters more equitable access to climate resilience planning tools and supports the broader societal goal of inclusive and effective climate adaptation. The anticipated outcomes of this research include the reduction of economic losses associated with climate change and the promotion of safer, more resilient communities. Through this work, we aim to contribute to the development of next-generation climate services that empower diverse stakeholders to engage meaningfully in climate action, regardless of their prior technical expertise.

Our approach is globally transferable: it adapts across countries by referencing local climate and socio-economic contexts without the expense of fine-tuning. In particular, climate-vulnerable developing countries stand to gain from region-specific climate services tailored to their needs. Effective deployment requires that the LLM incorporate not only country- and region-specific climate and hazard information but also knowledge of local industries, policy environments, and cultural contexts. Fine-tuning to encode such knowledge is often impractical due to the costs of data preparation and computation. Instead, a feasible path is to use the present model as a foundation model that captures common knowledge in climate science and adopt RAG that consults external information; in practice,

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this can be achieved simply by updating the document collection stored in a vector database. Experts on regional adaptation strategies should curate and validate the documents included in this vector database to ensure the relevance and reliability of region-specific knowledge. Most crucially, deployment hinges on access to regionally downscaled projections of future climate for target countries and regions. To establish the system as a standard tool, coordinated regional downscaling experiments across multiple regions are required.

To further enhance the capability of decision-support systems in highly interdisciplinary fields like climate services, future research must strategically evaluate the rapid architectural advancements in LLMs. Climate services inherently require diverse expertise, spanning climate science, economics, policy, industry, and culture. In the present study, we addressed this complexity by assigning multiple explicit roles to a single LLM to simulate structured deliberation. However, as generative AI evolves, there is growing interest in transitioning from this single-model approach to "multi-agent" frameworks. In such systems, each expert role is embodied by an autonomous AI agent capable of independent reasoning and decision-making. These agents dynamically interact and collaborate to provide comprehensive decision-making support. While this multi-agent approach offers significant potential, recent studies also highlight a crucial trade-off: when roles and instructions are rigorously defined, a single LLM can sometimes achieve higher accuracy and coherence than a complex network of interacting agents (Gao et al., 2025). Given these contrasting technological trends, it is essential to closely monitor developments in both single-LLM prompting and multi-agent architectures to determine the most effective approach for the highly nuanced demands of climate change adaptation.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Instruction tuning data set and LoRA adapter for the climatology-specific LLM are available on Zenodo at https://zenodo.org/records/18859658 (Matsuoka et al., 2026).

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Erratum

The originally published version of this article contained errors in the author contributions section. Coauthor D. Matsuoka contributed to the following: Conceptualization, Funding Acquisition, Methodology, Project Administration, and Writing – Original Draft. Coauthors S. Kawahara, K. Murakami, D. Sugiyama, M. Hayashida, K. Nguyen, A. Peng, T. Abe, and I. Sugiyama contributed to the role of Software. Coauthors R. Ito, S. Sugimoto, and M. Hara contributed to the role of Validation. This may be considered the authoritative version of record.

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